

Sukriti Gupta

LinkedIn://sukriti-gupta-iitd | Github://SukritiGupta | Google Scholar | Orchid id

EDUCATION

IIT DELHI

DOCTOR OF PHILOSOPHY IN COMPUTER SCIENCE AND ENGINEERING

July 2025 - Present

Advisors: Prof. Amit Kumar and

Prof. Naveen Garg

GPA: 9.571 / 10.0

IIT DELHI

B.TECH AND M.TECH IN COMPUTER SCIENCE AND ENGINEERING

Spl. in Theoretical Computer Science

July 2016 - May 2021

Advisor: Prof. Naveen Garg

GPA: 9.306 / 10.0

MONTFORT SCHOOL

Class XII-May 2016 | Delhi, India

Marks: 99.4% | CBSE

Class X-May 2014 | Delhi, India

GPA: 10.0 / 10.0 | CBSE

GRADUATE COURSEWORK

Advanced Algorithms

Algorithmic Graph Theory

Online Algorithms

Game Theory

Logic for Computer Science

Optimization

Numerical Algorithms

Cloud Computing

Parallel & Distributed Computing

Machine Learning

Artificial Intelligence

Foundations of Automatic Verification*

CO-CURRICULARS

•Vice Captain IITD Swimming Team (Apr 2017 - Mar 2018)

•Media and Comm. Manager, ShARE IITD (Jan 2018 - Jan 2019)

•Winner, National Aerolympics, 2014

VOLUNTEERING

• Teaching Mathematics and DSA to young girls from underprivileged backgrounds, initially through **SheCodes Foundation**, now on adhoc basis

• Working with visually impaired students to help them with **career guidance**, specifically towards JEE counselling

•Mentor for 7 first year students - IITD (Jan 2019 - May 2019)

PUBLICATIONS

APPROXIMATING OPTIMAL BROADCAST OF FILES IN A HOSE-MODEL NETWORK

Foundations of Software Technology and Theoretical Computer Science (FSTTCS) 2025 | Paper

- We solve the problem of minimizing makespan for file sharing between nodes having arbitrary upload and download capacities in a hose-model network under migratory and non-migratory settings
- Migratory model - using a novel integer program we show a schedule with $e^{1/e}$ OPT+P makespan; P is the time taken by slowest peer to download file.
- In the non-migratory model the problem is known to be NP-hard. We give a solution with makespan $18 \text{ OPT}+P$ and this is the first approximation algorithm for heterogeneous and asymmetric upload/download capacities.

AWARDS AND ACHIEVEMENTS

- Awarded **Outstanding TA** for Analysis and Design of Algorithms, Fall 2025
- Selected for **25th Estonian Winter School in Computer Science** (2020) with scholarship from Tallinn University of Technology and IIT Delhi.
- Awarded **IITD Merit Award** in sem 1, 2 & 6 for being in top 7% students
- Selected for exchange to **University Of Waterloo**, Canada for Fall 2018
- **All India CBSE Topper** for Class XII (2016): Scored the maximum marks amongst more than 1.2 million candidates
- Selected for **KVPY** Fellowship 2015 by Dept of Science and Technology
- Awarded **NTSE** (2014, NCERT) and **JTSE** (2012, DoE Delhi) scholarships

EXPERIENCE

ALPHAGREP SECURITIES

Quant Developer | Dec 2023 – Dec 2024 | Singapore

- Scaled index options market-making strats on NSE and expanded to BSE
- Researched alpha creation, signal generation, and market microstructure.

HFT Infra Developer | May 2021 – Dec 2023 | Bengaluru, India

- Specialized in low-latency C++ development for market data adapters and order routing systems for various crypto exchanges
- Analyzed exchange protocols & inefficiencies to gain trading edge

IIT DELHI

- Teaching assistant for NPTEL **Computational Geometry** (2020, 2021)
- Teaching assistant for **Analysis and Design of Algorithms** (Fall 2020, Fall 2025), **Theory of Computation** (Winter 2021)

TOWER RESEARCH CAPITAL | CORE ENGINEERING INTERN

May 2019 – July 2019 | Gurgaon, India

- Created a profiling tool which supported multi-threaded use cases
- Regression and benchmarking of proprietary tools, deployed setup on Jenkins.

PROCESSOR ARCHITECTURE LABORATORY EPFL

RESEARCH INTERN | PROF. PAOLO IENNE

May 2018 – July 2018 | Lausanne, Switzerland

- Passing information using custom pragma handling across a dynamic high level synthesis tool based on LLVM/CLANG framework.